Deliverable #1: Project Plan

Group #2 – Scrabble Game

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Introduction

This document contains all the documentation for the process activities we have performed and how we planned to accomplish our goals for this Scrabble-based project. Enclosed is our scope statement, org chart, Gantt chart sample, tools and standards documentation, configuration management plan, and weekly status reports.

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1.1 Scope Statement

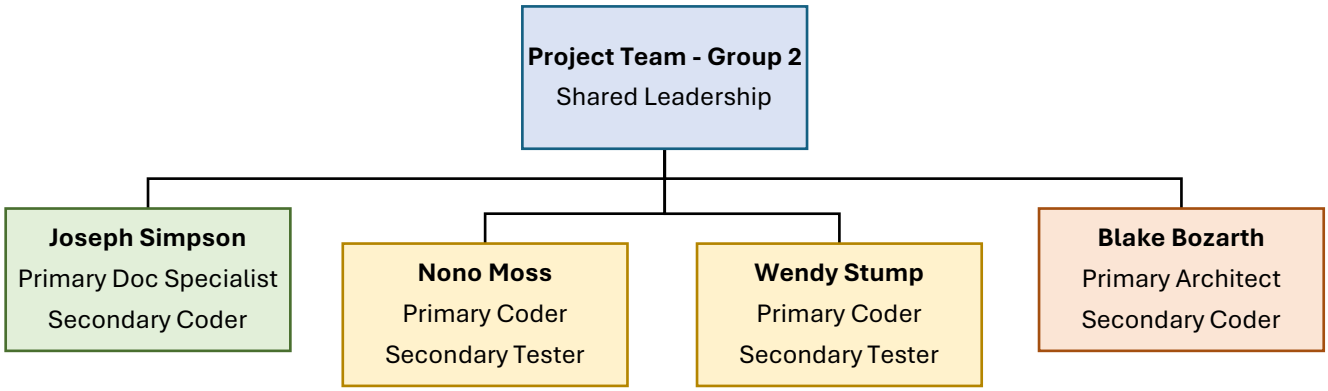
Group 2 will be developing a digital Scrabble-style game as our semester project. The core game mode will be a single-player, time-based game in which the user takes successive turns attempting to reach a target score before time expires. The required score and time limit will be determined by the selected difficulty level (easy, medium, or hard). All word placement rules, tile distribution, and board scoring will follow the standard rules of traditional Scrabble. The player will be provided with seven randomly drawn letter tiles and will draw new tiles after each turn to maintain a rack of seven tiles. The automatic word validation will be based on the SOWPODS dictionary.

If time permits, the following features will be implemented in the order listed:

- A two-player local pass-and-play mode following traditional Scrabble rules.
- A boss battler mode that introduces additional rogue-like mechanics layered onto the time-based single-player mode.
- A two-player mode against a computer-controlled opponent utilizing basic artificial intelligence.

The game will be implemented in Java using a graphical user interface built with JavaFX. The final application will be packaged as an executable JAR file, allowing it to run on Windows, macOS, and Linux systems with a compatible Java Runtime Environment installed. The software will function as a standalone application and will not require any additional software beyond the Java runtime.

1.2 Org Chart



1.3 Gantt Chart

Gantt.xlsx

1.4 Tools and Standards

1.4.1 Introduction

This document describes the various development tools and standards used throughout the project. These tools support implementation, collaboration, documentation, and testing.

1.4.2 Tools

- Programming Language
  - Java
- Integrated Development Environment (IDE)
  - Eclipse
- Build Management
  - Maven
- Version Control
  - GitHub
- User Interface (UI) Framework
  - JavaFX

1.4.3 Documentation Standards

- **Structure and Content**
  - Documentation must cover all items explicitly required in the project assignment PDF
  - Each section must provide sufficient detail such that a team unfamiliar with the project could understand, maintain, and extend the software.
- **Formatting**
  - Use Times New Roman, 12-point font.
  - Standard 1-inch margins on all sides.
  - 1.5 line spacing.
  - Include a linked Table of Contents for easy navigation through sections.
- **Traceability**
  - Each requirement must have a unique identifier (e.g., 1.2.0, 2.3.1) to allow tracing through design, implementation, and testing.
  - Source code and test cases must reference the requirement identifiers they satisfy.
- **Citations and References**
  - All external sources, standards, algorithms, or frameworks must be cited in APA format.
  - Include a complete References section at the end of each document.

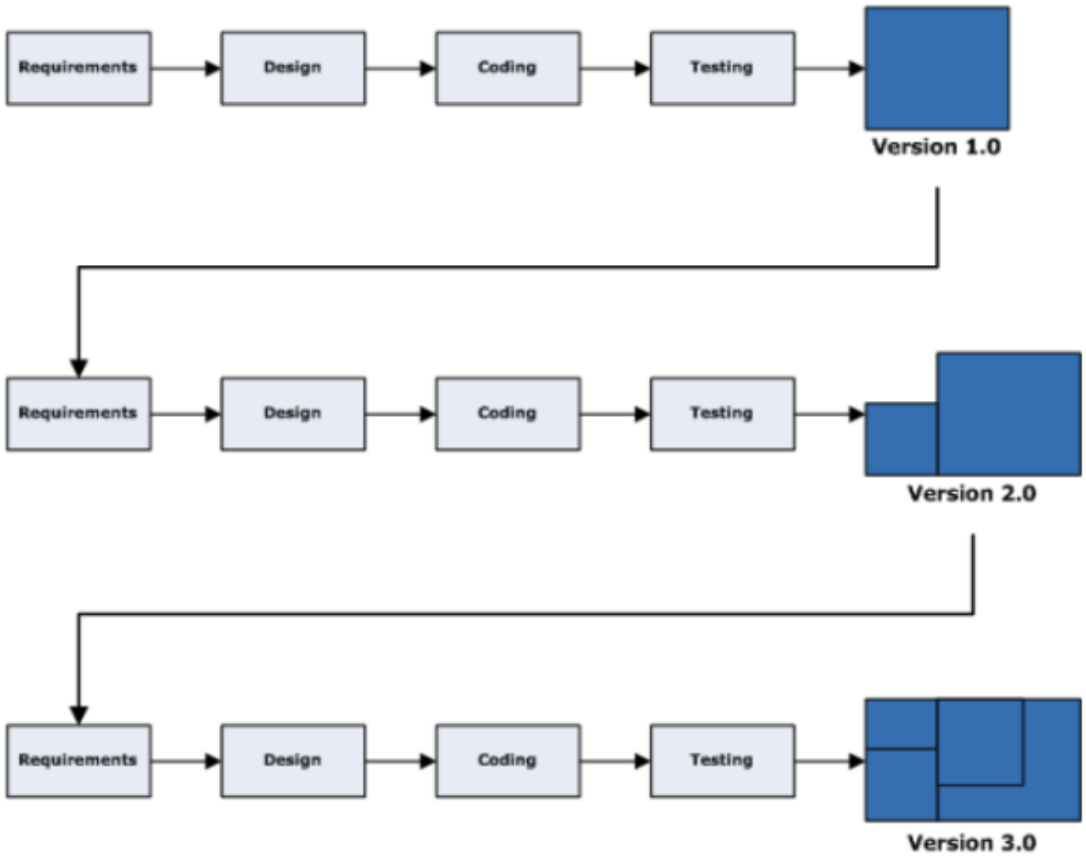
- **Professional Presentation**
  - Maintain consistent formatting, headings, and numbering throughout.
  - Figures, diagrams, tables, and screenshots should be clearly labeled and referenced in text.
  - Avoid typographical errors, incomplete sections, or placeholder content.
- **Document Completeness**
  - Documentation should allow a team unfamiliar with the project to understand the system requirements and build or maintain the software without further assistance.
  - All mandatory deliverables listed in the assignment PDF must be fully addressed.
- **Submission**
  - All documents will be submitted using Word unless otherwise specified.
  - Supporting diagrams, spreadsheets, or other files must be embedded or properly referenced.

1.4.4 Coding Standards

- **Structure and Organization**
  - Follow standard object-oriented design principles.
  - Each class must have a well-defined responsibility.
  - Packages must be organized by functionality.
- **Naming Conventions**
  - PascalCase for class names (ex. GroupProject)
  - camelCase for variables and methods (ex. completeProject())
- **Comments**
  - All public classes and methods must include Javadoc comments.

1.4.5 Process Model

Incremental (Iterative) Model:



The incremental model is an iterative approach where the system is developed and delivered in several increments, which include:

1. **Requirements**

- Define the requirements for the increment, including functionality features and constraints that apply. Requirements are to be reviewed before development begins.

2. **Design**

- Develop or design the defined features using Java or JavaFX. This step includes defining class structures, object relationships, or user interface layout. Design features are to be documented before implementation.

3. **Coding**

- Implement the defined features in Java. Maven is used to maintain project structure. Code is written following established Java coding standards to ensure readability and consistency throughout the project.

4. Testing

- Test the features to ensure it meets requirements. Any errors identified during testing will be corrected before completing the increment.

1.5 Configuration Management Plan

1.5.1 Overview

The project will use Git and GitHub to manage source code, track changes, and store release versions. The team will separate development versions from stable release versions using branches and version tags. This allows team members to work safely on new features while keeping stable working versions of the software available.

1.5.2 Tools Used

The project will use Git and GitHub to manage version control.

1.5.3 Branching Strategy

The repository will use three types of branches.

Main Branch

- Contains stable and working versions of the application
- Only completed and tested code will be merged here
- Each release version will be tagged on this branch

Development Branch

- Used for combining completed features and bug fixes
- May contain unfinished or partially tested code
- Completed and tested milestones will be merged to main from this branch

Feature Branches

- Created from the development branch
- Used for working on individual features
- Merged back into development when finished

1.5.4 Development Versions vs Release Versions

Development Versions:

Include feature branches and the development branch. These versions represent active work and may not be stable.

Release Versions:

Include code stored in the main branch and versions labeled with tags. These versions are stable and ready for use.

1.5.5 Version Numbering

The project will use the version format: **Major.Minor.Patch**

Major – Large project milestones and final release

Minor – New features added

Patch – Bug fixes or small improvements

1.5.6. Release Process

Features are completed in feature branches → Completed features are merged into the development branch → The development branch is tested and stabilized → Development is merged into the main branch → A version tag is created to finalize the release

1.5.7 Backup and Recovery

Git automatically keeps a complete history of all changes. If problems occur, the team can return to previous versions using commit history or version tags. Release tags will be used to mark working versions.

1.6 Weekly Status Reports

1.6.1 Week 1

1. **What was accomplished**

   This week, our group established our general idea for the project and created a Discord server to communicate more effectively. Specifically, we plan to create a digital Scrabble game using Java. Ideally, the game will have a pass-and-play local two-player mode and a single-player timed score-based mode with additional rouge-like boss battling elements if time allows. We have yet to pin down the exact format this will take; therefore, we will be submitting our scope tomorrow or Wednesday.

2. **How each group member contributed to the project**

   So far, every member has participated in group discussions, primarily providing our schedules, preferences, and ideas. Beyond that, individuals contributed the following:

   Blake – Drafting the general architecture for the project and outlined basic project requirements for the team.

   Nono – Created a Discord server for the team.

   Joseph – Wrote status report and doing initial research on Java Swing or JavaFX for GUI implementation

   Wendy – Created GitHub repository for the team, researched scrabble rules and the technical side of implementing a computer player / its feasibility with our project timetable vs. 2 local players or single player.

3. **What problems were encountered**

   We got off to a slightly slow start with establishing communication with the team but have since improved once Nono created the Discord server. Otherwise, scoping the project is going smoothly and only slightly delayed.

1.6.2 Week 2

1. **What was accomplished**

   This week, our group finished the Scope Statement and made solid progress working through Deliverable #1. Additionally, we began some very early work on drafting requirements, determined the roles for each team member, and decided on our group structure. We agreed that each member would be very flexible and ultimately chose a Democratic Decentralized team structure for the project. Individual roles are as follows:

   - Wendy: Primary Coder | Secondary Tester
   - Blake: Primary Architect | Secondary Coder
   - Nono: Primary Coder | Secondary Tester
   - Joseph: Primary Documentation Specialist | Secondary Coder
   - s

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily providing feedback for each other's work and ensuring that major decisions are discussed with the group.

   Blake – Finished the general architecture draft for the project, which ultimately drove our Scope Statement. Working through early draft of requirements.

   Nono – Assisting on drafting Deliverable #1.

   Joseph – Wrote Scope Statement and Status Report. Drafting overarching Word document for submission of Deliverable #1.

   Wendy – Created project Org Chart, finalized our GitHub Repository formatting, and created the Change Management Plan for our project.

3. **What problems were encountered**

   N/A

1.6.3 Week 3

1. **What was accomplished**

   This week, our group had our first live call to begin wrapping up most of Deliverable #1. We have nearly finished drafting the Gantt Chart and Tools and Standards documentation. Additionally, we continue to work on drafting our requirements documentation and some of the design documentation.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily discussing timelines and dividing out work this week.

   Blake – Working on drafting our requirements and design documentation.

   Nono – Drafted Tools and Standards documentation.

   Joseph – Wrote Status Report, beginning work on GUI implementation, and drafting Gantt Chart.

   Wendy – Assisted the whole team with using Maven and the GitHub repository and assisting with requirements documentation.

3. **What problems were encountered**

   N/A

1.6.4 Week 4

1. **What was accomplished**

   This week, our group completely finished our draft for Deliverable #1 (aside from the remaining status reports). Once all of the Deliverable #1 documents are combined and formatted into one Word document, we would like to send them to you for corrections. Additionally, we completed drafting the first two sections of our Requirements Documentation.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily giving feedback to one another.

   Blake – Continuing work on drafting our requirements and design documentation.

   Nono – Finished Tools and Standards documentation.

   Joseph – Wrote Status Report, finished Gantt Chart, working on combining and formatting Deliverable #1 documents.

   Wendy – Drafted the first two sections of our Requirements Documentation.

3. **What problems were encountered**

   N/A

1.6.5 Week 5

1. **What was accomplished**

   This week, we combined and formatted all of the Deliverable #1 documents (attached for your review). Additionally, we completed drafting our Specific Requirements section and began working on implementation.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions both over call and in chat, primarily giving feedback to one another.

   Blake – Completed Specific Requirements section.

   Nono – Working on implementation.

   Joseph – Wrote Status Report, combined and formatted Deliverable #1 documents, working on GUI implementation.

   Wendy – Working on implementation.

3. **What problems were encountered**

   Joseph has been placed on extended military orders to support national defense requirements. He is now working up to 12-hour days, 6-7 days a week for the near future. This may delay GUI implementation progress.

1.6.6 Week 6

1. **What was accomplished**

   This week, we continued working on implementation. Each member has taken their own section to work on and we have yet to combine our work. We expect to finish the first pieces of deliverable code by next Monday.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions both over call and in chat, primarily giving feedback to one another.

   Blake – Working on implementation.

   Nono – Working on implementation.

   Joseph – Wrote Status Report, working on GUI implementation.

   Wendy – Working on implementation.

3. **What problems were encountered**

   Multiple members had hectic work weeks, resulting in little progress being made compared to previous weeks. We believe we will be able to make some meaningful progress this week as Spring break has lightened our overall course loads.

1.6.7 Week 7

1. **What was accomplished**

   This week, we continued working on implementation. Each member has taken their own section to work on and we have begun sharing some draft work with each other.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily sharing code and giving feedback to one another.

   Blake – Working on implementation.

   Nono – Working on implementation.

   Joseph – Wrote Status Report, made Tile and BoardState class drafts (mostly just getters and setters), working on GUI implementation.

   Wendy – Working on implementation.

3. **What problems were encountered**

   N/A

1.6.8 Week 8

1. **What was accomplished**

   This week, we continued working on implementation. Each member has taken their own section to work on and we have begun uploading to our GitHub repository.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily sharing code and giving feedback to one another.

   Blake – Working on implementation.

   Nono – Working on implementation.

   Joseph – Wrote Status Report, working on GUI implementation.

   Wendy – Working on implementation, uploaded Java classes to repository.

3. **What problems were encountered**

   N/A

1.6.9 Week 9

1. **What was accomplished**

   This week, we continued working on implementation. Additionally, we have updated our GitHub repository to reflect our current progress and drafted more requirements documentation. Each member has taken their own section to work on and we have begun uploading to our GitHub repository.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily sharing code and giving feedback to one another.

   Blake – Working on implementation, updated architecture and requirements draft.

   Nono – Working on implementation.

   Joseph – Wrote Status Report, working on GUI implementation, uploaded documentation updates to repository.

   Wendy – Working on implementation, uploaded code updates to repository.

3. **What problems were encountered**

   N/A

1.6.10 Week 10

1. **What was accomplished**

   This week, we begin wrapping up implementation. So far we have drafted the following classes: Board, GameController, GameState, JavaFX, Main, PlacedTile, Rack, Tile, TileBag, TurnAnalyzer, WordValidation, and RulesEngine.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily sharing code and giving feedback to one another.

   Blake – Working on writing pseudocode for deliverables and wrapping up Deliverable #2.

   Nono – Updated TurnAnalyzer and GameController. Drafted RulesEngine.

   Joseph – Wrote Status Report, finished first draft of GUI implementation, including the board display, rack display, drag and drop, and other limited functions. Still need to handle continuous turns, updating scores, save states, and debugging.

   Wendy – Worked on bulk of other class implementation, uploaded updates to repository.

3. **What problems were encountered**

   N/A

1.6.12 Week 11

1. **What was accomplished**

   This week, we continued wrapping up implementation and are reviewing what is left in the project to complete. We are slightly behind schedule, but still plan to finish everything on time.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily sharing code and giving feedback to one another.

   Blake – Working on deliverables.

   Nono – Wrapping up final implementation.

   Joseph – Wrote Status Report, wrapping up GUI implementation.

   Wendy – Wrapping up final implementation.

3. **What problems were encountered**

   N/A

1.6.13 Week 12

1. **What was accomplished**

   This week, we began putting everything together for the final product, finishing initial implementation, debugging, collecting all of our documentation, and filling in any gaps we missed throughout the project.

2. **How each group member contributed to the project**

   So far, every member has meaningfully participated in group discussions, primarily sharing documents, giving feedback to one another, and discussing remaining work to be done.

   Blake – Working on Deliverable 2 finalization.

   Nono – Working on Deliverable 3 & 4 documentation.

   Joseph – Wrote Status Report, debugging.

   Wendy – Working on Deliverable 3 & 4 documentation.

3. **What problems were encountered**

   N/A